

Ayoung Park

Financial Management Stabilization Department, Financial Supervisory Service
Seoul 07321, Republic of Korea
+82-10-2100-1617
ayoungp@fss.ok.kr, ayppark@gmail.com

WORK EXPERIENCE

Senior Manager Jan 2026 - present
Supervisory Strategy Research Team, Supervision Innovation Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on blockchain-based financial networks, regulatory frameworks, and supervisory innovation (including AML and crypto-assets).

Senior Manager Mar 2024 - Jan 2026
Future Finance Research Team, Financial Market Stabilization Department, Financial Supervisory Service, Seoul, Republic of Korea

- Conducted research on Financial Market Analysis related with the financial stability, regulation, and innovation.

Research Assistant Professor July 2021 - Feb 2024
The Institute of Knowledge Management, Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Conducted research on Financial Market Analysis via Connectedness and Deep Learning with the 5-years funding from National Research Foundation(NRF) of Korea

Visiting Scholar Jan 2023 - Feb 2023
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the prediction of the speed of collapse and recovery in the 2007-2009 global financial crisis on international stock markets via complex network theory.

Lecturer Mar 2021 - June 2021
Division of Business Administration, Chosun University, Gwangju, Republic of Korea

- Played a pivotal role in the conceptualization and creation of the *Introduction to Fintech* course, ensuring its relevance and applicability in the current financial landscape.

Visiting Scholar Mar 2019 - Feb 2020
Department of Anesthesiology, University of Michigan Medical School, Ann Arbor, MI, United States

- Research on the general mechanism of response and recovery between anesthesia on the brain and the impact of the 2007-2009 global financial crisis on international stock markets.

Visiting Scholar Jan 2019 - Feb 2019
Department of Physics, Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea

- Research on the non-linear dynamics and complexity based on brain and financial market.

Visiting Scholar

Jan 2017 - Feb 2017

Division of Business Administration, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, United States

- Research on the role of board diversity and behavior finance.

Education**Ph.D. in Finance**

Feb 2021

Chosun University, Gwangju, Republic of Korea

- Dissertation title: Essays on Financial Connectedness, Systemic Risk, and Performance.
- Excellent Research Award

M.S. in Finance

Aug 2016

Chosun University, Gwangju, Republic of Korea

- Thesis title: The Empirical Study on Measuring Systemic Risk using Variance Decomposition and Debrank method

B.Sc. in Mathematics and B.A. in Business Administration

Feb 2015

Chosun University, Gwangju, Republic of Korea

- Summa Cum Laude

PROFESSIONAL TRAINING

Anti-Money Laundering(AML) Specialist Program, Hanyang University, Dec 2025 – Focus on AML regulation, crypto-assets, and financial surveillance

RESEARCH INTERESTS

Computational Finance, Risk Management, Financial Intermediation, Empirical Asset Pricing, Corporate Governance, Complex System, Graph Neural Network, Natural Language Processing

RESEARCH FUNDING

Title: *Financial Market Analysis via Connectedness and Deep Learning.*

- **Objective:** Investigating the connectedness among financial institutions and boards of directors from international countries. The focus lies on risk propagation, knowledge spillover, and its predictive potential for technological innovation.
- **Source:** National Research Foundation of Korea (NRF).
- **Amount:** A grant of 200 million KRW.
- **Duration:** July 2021 - Mar 2024.

WORKING PAPERS

- **Global Banking Networks and Loan Pricing**, Job Market Paper
This study examines the influence of the core-periphery structure in interbank networks on the cost of credit in syndicated loans. Utilizing syndicated loan data from 2000 to 2020 across 39 countries, we construct monthly interbank networks, identifying core and peripheral banks. Our findings reveal a significant correlation between a bank's position in the network core and increased volatility during periods like the 2007-2008 financial crisis and the COVID-19 pandemic, indicating a higher susceptibility of core banks to systemic risks. Additionally, we observe that banks at the network's core tend to charge higher interest rates on loans, suggesting an association between network centrality and loan pricing. This research provides insights into the role of interbank network structures in financial stability and risk management.

- **Best Paper Award, Semi-finalist in Financial Intermediation & Markets at Financial Management Association (FMA) 2023**

- **Best Student Paper Award** at Fall conference of Korea Academy of Complexity Studies 2018 with a prize of 200,000 KRW (approx. 181 USD)
- **Computing in Economics and Finance (CEF) 2023**
- **Workshop of Asia- Pacific Applied Economics Association, 2023**
- **Stablecoin Prices as Shadow FX Quotes: Dollar News and the Korean Crypto Premium**
Scheduled for presentation at the Financial Management Association Annual Meeting, 2026.
- **Beyond AUC: Evaluating Model Warnings under Limited Review Capacity: Evidence from the New York Fed Market Participant Aggregate Path**
- **Forecast Compression and Borrower-Size Gradients in Syndicated Lending**
- **Holdings Embeddings and Subsequent Illiquidity**
- **Missing Carbon and Model Risk**
- **Predictable, Not Deployable: Economic Networks and the Transportability of Housing Forecasts**
- **Investor Sentiment and Stock Return Predictability: The Influence of Online Stock Discussion Boards**, (with Danbi Park, Gabjin Oh)
In this study, we investigate the impact of investor sentiment, derived from social media activities on platforms like Naver’s online stock discussion board, a leading search engine in South Korea. This sentiment is particularly noteworthy given the significant role of retail investors in the South Korean stock market. We analyze expressions of bullishness and disagreement in these forums to assess their correlation with retail investor activity. Our findings reveal that social media sentiment markedly enhances the predictability of stock returns and market volatility in companies with high retail trading activity, particularly in firms heavily traded by retail investors. This underscores the critical role of social media in influencing retail investment strategies in South Korea’s unique stock market environment.

JOURNAL ARTICLES

See also [my google scholar](#) page.

- Uncheol Lee, Hyoungkyu Kim, Minkyung Kim, Gabjin Oh, Pangu Joo, Ayoun Park, Jamie Sleight, Katie Warnaby, Iren Tracey, George Mashour, [Proximity to Explosive Synchronization Determines Network Collapse and Recovery Trajectories in Neural and Economic Crises](#), *PNAS*, (SCI, impact factor: 9.1, Ranked 14/135 in Multidisciplinary Sciences), Sep 2025.
- Ayoun Park and Gabjin Oh, [The Contagion Effect of Heterogeneous Investor Groups](#), *Plos One* (SCI, impact factor: 3.752, Ranked 29/134 in Multidisciplinary Sciences), Oct 2023.
- Ayoun Park and Gabjin Oh, [The Core of the Board Networks and Firm Value](#), *Frontiers in Physics* (SCI, impact factor: 3.718, Ranked 42/111 in Physics, Multidisciplinary), Feb 2023.
- Ayoun Park and Gabjin Oh, [Economic System Research by Using Statistical Physics](#), *Physics & High technology* (Highlighted Article), Aug 2022.

- Jinjoo Yoon, Ayoung Park, and Gabjin Oh, [Grouping Analysis of Social Media in Korean Financial Market](#), *New Physics: Sae Mulli* (SCOPUS), Feb 2022.
- Ayoung Park and Gabjin Oh, [Lending Diversification and Interconnectedness of Syndicated Loan Market](#), *Frontiers in Physics* (SCI, impact factor: 3.56, Ranked 30/110 in Physics, Multidisciplinary), Jan 2021.
- Jie Deng, A-young Park, and Gabjin Oh, [The Relationship between Cash Flows and Investment according to Economic Status](#), *Journal of the Korean Physical Society* (SCI, impact factor: 0.649, Ranked 87/111 in Physics, Multidisciplinary), Jan 2020.
- Ayoung Park and Gabjin Oh, [Systemic risk and Expected return](#), *New Physics: Sae Mulli* (SCOPUS), June. 2018.
- Ayoung Park and Gabjin Oh, [An Analysis of the Relationship between Market Risk and Information Flows among Trading Volume of the Industry Sector in the Korean Stock Market](#), *New Physics: Sae Mulli* (SCOPUS), May. 2017.
- Gabjin Oh, Ho-Yong Kim, and Ayoung Park, [Analysis of technological innovation based on citation information](#), *Quality and Quantity* (SSCI, impact factor: 1.25, Ranked 42/96 in Social Sciences, Interdisciplinary), Dec. 2016.
- A Young Park, Ho-Yong Kim, and Gabjin OH, [An Empirical Study on Measuring Systemic Risk Based on Information Flows using Variance Decomposition and DebtRank](#), *The Korean Operations Research and Management Science Society*, (KCI), Nov. 2015.

WORK IN PROGRESS

- Board Diversity and CEO Power
 - **Best Student Paper Award** at Fall conference of Korea Academy of Complexity Studies 2017 with a prize of 500,000 KRW (approx. 450 USD)
- A Network Approach to Optimal Portfolio

TEACHING EXPERIENCE

Introduction to Fintech, undergraduate, Chosun University (Spring 2021-2023)
Security Investment Practice, undergraduate, Chosun University (Spring 2023)
Management Statistics, undergraduate, Chosun University (Fall 2021, 2023)
Management and Economics, undergraduate, Chosun University (Fall 2021-2023)

SPECIAL ACHIEVEMENTS

Awards

- *Best Paper Award*, Semi-finalist in Financial Intermediation at Financial Management Association (FMA), 2023
- *Ph.D. Excellent Research Award*, Chosun University, 2021
- *Best Research Award* for "Measuring Information Asymmetry of Interbank Network from Syndicated Loan Market", Fall Conference of Korea Academy of Complexity Studies, 2018
- *Best Research Award* for "Diversity in the Board Network Impact on Firm Performance", Fall Conference of Korea Academy of Complexity Studies, 2017
- *Encouragement Award* for "Analysis of Financial Stability Based on Investor Trading Behavior", 4th Conference of Social Sciences Korea Networking, 2017
- *Fellowship*, Brain Korea 21+, Korean Government, 2015-2020

- *Summa Cum Laude*, Department of Mathematics, Chosun University, 2014
- *Scholarship*, Catholic Archdiocese of Gwangju, 2014
- *Scholarship*, KT&G Scholarship Foundation, 2013-2014
- *Full-Tuition Scholarship*, Korea Student Aid Foundation (KOSAF), 2011-2014

Industry Experience

- *Researchers*, Technological Innovation and Data Construction for Financial/Economic Digital Twins in Companies and Industries at ETRI(Electronics and Telecommunications Research Institute), 2022- 2023

Volunteer Experience

- *Global Volunteer*, Happy Move funded Hyundai and Habitat: Participated in community house-building in the Philippines and intercultural exchange, Feb 2014
- *KT&G 7st SangSang Univ*, University Student Planning Group, 2013
- *Education*, Elementary school in Catholic, 2011-2014.

Invited Lectures

- Global Banking Network and Syndicated Loan Prices, at workshop of Asia-Pacific Applied Economics Association, Xian Jiaotong University, July. 2023
- Network Analysis in the Financial System, Invited Talk at conference of Korea Academy of Complexity Studies, Sookmyung Women's University, Dec. 2022; at the Ajou University, Oct. 2022; at the Pukyong National University, Aug. 2022
- From Research to Regulatory Affairs, Invited Talk at Inha University, Nov. 2025

PROFESSIONAL INTERVIEWS

- Invited Interview in [National Research Foundation of Korea\(NRF\)](#), Oct 2021 (based on the high score of research project)

CONFERENCE CONTRIBUTIONS

- Global Banking Network and Syndicated Loan Prices, *Sydney Banking and Finance Conference (SBFC) 2025*, *2023 Financial Management Association (FMA)*, *29th International Conference Computing in Economics and Finance (CEF)*, *2023 Fall Conference of Korean Finance Association*, *15th Asia Pacific Physics Conference*, *24th Interdisciplinary KASBA conference*, *Networks 2021: A Joint Sunbelt and NetSci Conference*, *2019 The Network Science Society*
- A Network Approach to Optimal Portfolio: Application to the Korean Stock Market, *29th International Conference Computing in Economics and Finance (CEF)*
- Measuring Similarity between trend Behaviors of Multivariate Time Series, *2023 European Financial Management Association (EFMA)*, *28th International Conference Computing in Economics and Finance (CEF)*
- Evolution of Social media activity-financial market Based on the complex network method, *28th International Conference on Statistical Physics*
- The Effect of Knowledge Diversity of Board Networks on Corporate Value, *27th International Conference Computing in Economics and Finance (CEF)*,

2018 Econophysics Colloquium, 2018 Asia Pacific Econophysics Conference, 2017 Fall conference of Korea Academy of Complexity Studies

- Analysis of Mutual Fund Similarity Network in Korea Equity Market, *2018 Asia Pacific Econophysics Conference, The 13th INFINITI Conference on International Finance*
- Financial Stability Based on Information Flows among Investor Trading Behavior, *25th International Conference Computing in Economics and Finance (CEF), 2018 Asia Pacific Econophysics Conference, NetworkX-2018: International School and Conference on Network Science, 2017 International Conference on SigmaPhi, 2017 4th conference of Social Sciences Korea Networking (SSK), 2016 Japanese Association of Financial Econometrics and Engineering (JAFEE)*
 - Encouragement student paper award at the 4th conference of Social Sciences Korea Networking (SSK).
- Usefulness of the Fama and French three-factor model in Shanghai A-share market, *2016 International School and Conference on Network Science, 2016 Fall conference of Korea Academy of Complexity Studies*

PROFESSIONAL SERVICE

- **Journal Referee:** Plos One, Journal of the Korea Society Industrial Information System, Humanities and Social Sciences Communications
- **Conference Discussant:** Fall conference of Korean Finance Association 2022; European Financial Management Association (EFMA) 2023; Financial Management Association (FMA) 2023, Asia Financial Management Association (Asia FMA) 2024
- **Conference Reviewer:** Financial Management Association (FMA) 2023
- **Conference Session Chair:** Computing in Economics and Finance (CEF) 2022, 2023; Financial Management Association (FMA) 2023; Asia Financial Management Association (Asia FMA) 2024
- **Professional Membership:** Financial Management Association (FMA), Korean Security Association, Korea-America Finance Association(KAFA)

SKILLS

Python, Stata, Matlab, C++, Latex, Gephi

LANGUAGES

English (Fluent), Korean (Native)

REFERENCE

- **Gabjin Oh, Ph.D.**, Professor, Division of Business Administration, Chosun University, Gwangju, Republic of Korea, *phecogjoh@chosun.ac.kr*
- **Uncheol Lee, Ph.D.**, Associate Professor, Department of Anesthesiology, Associate Director, Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI, United States, *uclee@med.umich.edu*
- **Woo-Sung Jung, Ph.D.**, Professor, Department of Industrial and Management Engineering/Physics, Pohang University of Science and Technology (POSTECH), Pohang, Republic of Korea, *wsjung@postech.ac.kr*
- **O-Kyu Kwon, Ph.D.**, Senior Researcher, NIMS-National Institute for Mathematical Sciences, South Korea, *okyu.kwon@gmail.com*
- **Hyeng Keun Koo, Ph.D.**, Professor, Department of Financial Engineering, Ajou University, Suwon, Republic of Korea, *hkoo@ajou.ac.kr*